Individual Assessment Coversheet

To be attached to the front of the assessment.

Campus: Online

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Group: Group 1

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Indicate	Yes	No
Plagiarism report attached	✓	

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Question 1

client.py

```
""" CLIENT SIDE """

import socket


SERVER_IP = '127.0.0.1'

PORT = 5500


def start_client():

    client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)


    print("=" * 40)

    print("WMS Client")

    print("=" * 40)


    try:

        print(f"Connecting to {SERVER_IP}:{PORT}...")

        client_socket.connect((SERVER_IP, PORT))

        print("Connected to server\n")


        while True:

            print(f"What item are you looking for? ('exit' to quit)")

            message = input()


            if message.strip().lower() == "exit":

                print("Disconencting from server")

                client_socket.sendall("Exit".encode('utf-8'))
```

```
            break

        if message:
            client_socket.sendall(message.encode('utf-8'))

            reply = client_socket.recv(1024).decode('utf-8')
            print(reply, end='')

    except ConnectionRefusedError:
        print(f"[ERROR] Could not connect to server at {SERVER_IP}:{PORT}")

    except Exception as e:
        print(f" Error: {e}")

    finally:
        # Close connection
        client_socket.close()
        print("[DISCONNECT] Successful\n")

if __name__ == "__main__":
    start_client()
```

WMS_Server.py

```
""" SERVER SIDE """
import socket

SERVER_IP = '127.0.0.1'
PORT = 5500
```

```
server_running = True

items = {
    "Nails": 200,
    "Screws": 321,
    "Toolkit": 59,
    "Wood Planks": 97,
    "Sheet Metal": 153,
    "Lightbulb": 76,
}

def handle_client(client_socket, client_address):
    global server_running

    try:
        print(f"Client {client_address} connected")
        while True:
            data = client_socket.recv(1024).decode('utf-8').strip()

            # in case client presses enter
            if not data:
                continue

            # Client Exit Code
            if data.lower().strip() == "exit":
                print(f" [DISCONNECT] {client_address} sent the exit code.")
                server_running = False
```

```
            break

        # search through dictionary to find item
        if data in items and items[data] > 0:
            message = f"[ITEM FOUND] {data} : {items[data]}"
        else:
            message = f"[ITEM NOT FOUND] {data} is not in stock"

        client_socket.sendall((f"{message}\n").encode('utf-8'))
        print(f"{message}\n")

    except ConnectionResetError:
        print(f"Error: Client Closed Connection Upbruptly")
    except Exception as e:
        print(f"Error: {e}")



def start_server():
    """Start TCP Server"""

    server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    server_socket.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)

    server_socket.bind((SERVER_IP, PORT))

    server_socket.listen()
    global server_running
```

```
# Allows for keyboard interrupt on server side every 1 second
server_socket.settimeout(1.0)

print(f'=' * 40)
print(" TCP WMS Server Started")
print("=" * 40)
print(f" Listening on {SERVER_IP}:{PORT}")

try:
    while server_running == True:

        try:
            client_socket, client_address = server_socket.accept()

            # When client connects, remove timeout
            client_socket.settimeout(None)

            handle_client(client_socket, client_address)

            if server_running == False: break

        except socket.timeout:
            continue

except KeyboardInterrupt:
    print(f"\n\n Server interrupted by keyboard input")

finally:
```

```
    if 'client_socket' in locals():
        client_socket.close()
    server_socket.close()


    print(f" Server Closed\n")


if __name__ == "__main__":
    start_server()
```

OUTPUT

```
PS C:\Users\lkren\OneDrive\Documents\ITPNA2-T12_Projects\ITPNA2-12_FormativeAssessmen
t_Online_EDUV7579091\q1> py .\WMS_Server.py
========================================
 TCP WMS Server Started
========================================
 Listening on 127.0.0.1:5500
Client ('127.0.0.1', 62946) connected
[ITEM FOUND] Nails : 200

[ITEM FOUND] Screws : 321

[ITEM NOT FOUND] Chairs is not in stock

 [DISCONNECT] ('127.0.0.1', 62946) sent the exit code.
 Server Closed

PS C:\Users\lkren\OneDrive\Documents\ITPNA2-T12_Projects\ITPNA2-12_FormativeAssessmen
t_Online_EDUV7579091\q1>
```

```
PS C:\Users\lkren\OneDrive\Documents\ITPNA2-T12_Projects\ITPNA2-12_FormativeAssessmen
t_Online_EDUV7579091\q1> py .\client.py
========================================
WMS Client
========================================
Connecting to 127.0.0.1:5500...
Connected to server

What item are you looking for? ('exit' to quit)
Nails
[ITEM FOUND] Nails : 200
What item are you looking for? ('exit' to quit)
Screws
[ITEM FOUND] Screws : 321
What item are you looking for? ('exit' to quit)
Chairs
[ITEM NOT FOUND] Chairs is not in stock
What item are you looking for? ('exit' to quit)
exit
Disconencting from server
[DISCONNECT] Successful

PS C:\Users\lkren\OneDrive\Documents\ITPNA2-T12_Projects\ITPNA2-12_FormativeAssessmen
t_Online_EDUV7579091\q1>
```

Question 2

bookings.py

```
# QUESTION 2

from operator import itemgetter

client_data = []

FIXED_PRICE = 20


def handle_input():

    print("Client Name:")

    c_name = input()


    print("\nDestinaion:")

    destination = input()


    print("\nNumber of Passengers:")

    passengers = input()


    write_ln = f"{c_name} | {destination} | {passengers}\n"


    return write_ln


def handle_file(write_ln):

    with open("client_info.txt", "a+") as file:

        file.write(write_ln)


def order_data():

    with open("client_info.txt", "r") as file:
```

```
        for line in file:
            data = line.strip().split('|')
            if data:
                client_data.append(data)

    for clients in client_data:
        total_price = clients[2]
        total_price = total_price.lstrip()
        total_price = int(total_price)

        total_price = total_price * FIXED_PRICE
        clients.append(total_price)

    client_data.sort(key=itemgetter(3), reverse=True)

    print(f"=" * 70)
    print(f"{' NAME':<20}{'DESTINATION':<20}{'PASSENGERS':<20} PRICE")
    print("=" * 70)
    for client in client_data:
        output = f" {client[0]:<20}{client[1]:<24}{client[2]:<15} ${client[3]}\n"
        print(output)


def start_booking():

    while True:
        print(f"What would you like to do?")
```

```
    print("(A) Create a booking\n(B) View Current Bookings\n(C) Exit Application\n")
    selection = input().upper()

    if selection == 'A':
        write_ln = handle_input()
        handle_file(write_ln)
        print("Client Added Successfully\n")
    elif selection == 'B':
        order_data()
    elif selection == 'C':
        break
    else:
        print("Please Enter a Valid Option\n")


if __name__ == "__main__":
    start_booking()
    print("Closing Application")
```

OUTPUT

```
t_Online_EDUV7579091\q2> py .\bookings.py
What would you like to do?
(A) Create a booking
(B) View Current Bookings
(C) Exit Application

A
Client Name:
John Peters

Destinaion:
Paris, France

Number of Passengers:
3
Client Added Successfully

What would you like to do?
(A) Create a booking
(B) View Current Bookings
(C) Exit Application

B
=========================================================================
 NAME                DESTINATION          PASSENGERS           PRICE
=========================================================================
 Susan                 Chicago                 7              $140

 Steve                 Paris                   5              $100

 John Peters           Paris, France           3              $60

 John                  Cape Town               2              $40

 Tom                   Sydney                  1              $20

What would you like to do?
(A) Create a booking
(B) View Current Bookings
(C) Exit Application

C
Closing Application
```

Question 3

generate_certificates.sh

```
#!/bin/bash

# Generate self-signed certificate for testing


# DO NOT use self-signed certificates in production!



echo "Generating SSL certificate for testing..."



openssl req -x509 -newkey rsa:4096 -nodes -keyout server-key.pem -out server-cert.pem -days 365 -subj "//CN=localhost"


echo "Certificate generated successfully!"


echo "Files created:"


echo "  - server-key.pem (private key)"


echo "  - server-cert.pem (certificate)"
```

tls_client.py

```
import socket

import ssl


SERVER_IP = 'localhost'

PORT = 8443

SERVER_CERT = 'server-cert.pem'


def create_tls_client():

    context = ssl.create_default_context(ssl.Purpose.SERVER_AUTH)


    # Prevent SSL error without disabling the security checks

    context.load_verify_locations(SERVER_CERT)


    client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)


    tls_socket = context.wrap_socket(client_socket, server_hostname=SERVER_IP)


    print(f"[CONNECTING] Connecting to {SERVER_IP}:{PORT}")

    tls_socket.connect((SERVER_IP, PORT))

    print("[SUCCESS] Connected Successfully")


    return tls_socket



def start_client():

    print("=" * 60)

    print("TLS/SSL Client")
```

```
    print("=" * 60)

    try:
        tls_socket = create_tls_client()

        cipher = tls_socket.cipher()
        if cipher:
            print("[SUCCESS] TLS Connection Established:")
            print(f"\tCipher {cipher[0]}\n\tProtocol: {cipher[1]}\n\tKey Bits: {cipher[2]}")

            message = "Hello from Secure Client ^^"
            print(f"\n[SENDING...] '{message}'")
            tls_socket.sendall(message.encode('utf-8'))

            response = tls_socket.recv(4096)
            print(f"[RECEIVED] Response from Server: '{response.decode('utf-8')}'")

            tls_socket.close()
            print(f"[CLOSED] TLS Socket Closeed Successfully!")

    except Exception as e:
        print(f"[ERROR] Could not close Connection\n{e}")
        print("[TIP] Make sure TLS Server is running")

if __name__ == "__main__":
    start_client()
```

tls_server.py

```
import socket

import ssl


SERVER = '127.0.0.1'

PORT = 8443

CERT_FILE = 'server-cert.pem'

KEY_FILE = 'server-key.pem'


def create_server():

    context = ssl.SSLContext(ssl.PROTOCOL_TLS_SERVER)


    try:

        context.load_cert_chain(certfile=CERT_FILE, keyfile=KEY_FILE)

    except:

        print("[ERROR] Certificate files not found")

        print("Run generate_certificates.sh first")

        exit(1)


    context.minimum_version = ssl.TLSVersion.TLSv1_2


    server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

    server_socket.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)


    server_socket.bind((SERVER, PORT))


    server_socket.listen(5)
```

```
    server_socket.settimeout(1.0)

    return server_socket, context


def handle_client(client_socket, client_address):
    print(f"\nConnection from {client_address[0]}:{client_address[1]}")

    cipher = client_socket.cipher()
    if cipher:
        print(f"\tCipher: {cipher[0]}")
        print(f"\tTLS Version {cipher[1]}")

    try:
        data = client_socket.recv(4096)

        if data:
            message = data.decode('utf-8')
            print(f"Received Message: {message}")

            response = f"Server Received: {message}"
            client_socket.sendall(response.encode('utf-8'))
            print(f"\tSent Response")

    except Exception as e:
        print(f"[ERROR] Client Handle Error: {e}")

    finally:
```

```
        client_socket.close()
        print(f"\tClient {client_address[0]} closed")



def start_server():
    print("=" * 60)
    print(f"TLS/SSL Server")
    print("=" * 60)
    print(f"Starting TLS/SSL Server on {SERVER}:{PORT}")
    print(f"Certificate: {CERT_FILE}")
    print(f"Private Key: {KEY_FILE}")

    server, context= create_server()

    try:
        while True:
            try:
                raw_client_socket, client_address = server.accept()
                secure_client_socket = context.wrap_socket(raw_client_socket,
server_side=True)
                handle_client(secure_client_socket, client_address)
                raw_client_socket.close()

            except socket.timeout:
                continue

    except ssl.SSLError as e:
        print(f"[SECURITY ALERT] Handshake failed for {client_address[0]}")
```

```
        print(f"Clear failure message: Unauthorized or bad certificate. Details: {e}")
        #raw_client_socket.close()

    except Exception as e:
        print(f"[ERROR] SERVER ERROR: {e}")

    except KeyboardInterrupt:
        print("\n\nServer Stopped by user")

    finally:
        raw_client_socket.close()
        server.close()
        print("Server Socket Closed")


if __name__ == "__main__":
    start_server()
```

OUTPUT

```
PS C:\Users\lkren\OneDrive\Documents\ITPNA2-T12_Projects\ITPNA2-12_FormativeAssessmen
t_Online_EDUV7579091\q3> py .\tls_server.py
============================================================
TLS/SSL Server
============================================================
Starting TLS/SSL Server on 127.0.0.1:8443
Certificate: server-cert.pem
Private Key: server-key.pem

Connection from 127.0.0.1:54578
        Cipher: TLS_AES_256_GCM_SHA384
        TLS Version TLSv1.3
Received Message: Hello from Secure Client ^^
        Sent Response
        Client 127.0.0.1 closed


Server Stopped by user
Server Socket Closed
PS C:\Users\lkren\OneDrive\Documents\ITPNA2-T12_Projects\ITPNA2-12_FormativeAssessmen
t_Online_EDUV7579091\q3>
```

```
PS C:\Users\lkren\OneDrive\Documents\ITPNA2-T12_Projects\ITPNA2-12_FormativeAssessmen
t_Online_EDUV7579091\q3> py .\tls_client.py
============================================================
TLS/SSL Client
============================================================
[CONNECTING] Connecting to localhost:8443
[SUCCESS] Connected Successfully
[SUCCESS] TLS Connection Established:
        Cipher TLS_AES_256_GCM_SHA384
        Protocol: TLSv1.3
        Key Bits: 256

[SENDING...] 'Hello from Secure Client ^^'
[RECEIVED] Response from Server: 'Server Received: Hello from Secure Client ^^'
[CLOSED] TLS Socket Closeed Successfully!
PS C:\Users\lkren\OneDrive\Documents\ITPNA2-T12_Projects\ITPNA2-12_FormativeAssessmen
t_Online_EDUV7579091\q3>
```

Question 4

client.py

```
import socket

import os



SERVER_IP = '127.0.0.1'

SERVER_PORT = 5001


FILE_STORE = 'client_files'



def send_files(file_path, client_socket):

    try:

        file_path = os.path.join(FILE_STORE, file_path)

        if not os.path.exists(file_path):

            print(f"[ERROR] File not found: {file_path}")

            return


        file_name = os.path.basename(file_path)

        file_size = os.path.getsize(file_path)


        print(f"[SENDING...] {file_name} to {SERVER_IP}")

        client_socket.sendall(file_name.encode('utf-8'))


        print(f"[SENDING...] {file_size} bytes to {SERVER_IP}")

        client_socket.sendall(str(file_size).encode('utf-8'))
```

```
        bytes_sent = 0
        with open(file_path, 'rb') as file:
            while bytes_sent < file_size:
                chunk = file.read(4096)
                if not chunk:
                    break

                client_socket.sendall(chunk)
                bytes_sent += len(chunk)
                progress = (bytes_sent / file_size) * 100
                print(f"\rProgress: {progress:.1f}%", end='', flush=True)

            print(f"\rProgress: 100.0%")
            print(f" Sent {bytes_sent:,} bytes\n")
    except Exception as e:
        print(f"[ERROR] Send file: {e}")



def req_files(file_name, client_socket):
    try:
        print(f"[REQUEST] {file_name} from {SERVER_IP}")
        client_socket.sendall(file_name.encode('utf-8'))

        file_size = int(client_socket.recv(1024).decode('utf-8'))
        print(f"\tSize: {file_size:,} bytes ({file_size/1024:.2f} KB)")

        file_path = os.path.join(FILE_STORE, file_name)
```

```
bytes_received = 0

with open(file_path, 'wb') as file:
    print(f"   Receiving data...", end='', flush=True)

    while bytes_received < file_size:
        bytes_to_read = min(4096, file_size - bytes_received)
        chunk = client_socket.recv(bytes_to_read)

        if not chunk:
            print("\nNOT CHUNK")
            break

        file.write(chunk)
        bytes_received += len(chunk)
        if bytes_received % 102400 == 0:
            progress = (bytes_received / file_size) * 100
            print(f"\r   Progress: {progress:.1f}%", end='', flush=True)

print(f"\r   Progress: 100.0%")

if bytes_received == file_size:
    print(f"[SAVED] {file_path}")
    print(f"[COMPLETE] {bytes_received:,} bytes\n")
else:
    print(f"[INCOMPLETE] File Transfer Incomplete {bytes_received}/{file_size}")
```

```
    except Exception as e:
        print(f"[ERROR] Cannot Receive File\n{e}")




def start_client():
    #Start client side
    client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

    print("=" * 60)
    print("Multi-Client File Transfer Server Started")
    print("=" * 60)

    try:
        print(f"Connecting to {SERVER_IP}:{SERVER_PORT}")
        client_socket.connect((SERVER_IP, SERVER_PORT))
        print("Connected to Server\n")

        print(f"Please Enter your Name:\n")
        name = input()
        client_socket.sendall(name.encode('utf-8'))

        while True:
            print("Would you like to send or request a file?")
            print(" (1) SEND FILE\n (2) REQUEST FILE\n (3) EXIT PROGRAM\n")
            selection = input()
```

```
            client_socket.sendall(selection.encode('utf-8'))

            if selection.strip() == '1':
                print("Please enter file path of the file you want to send:")
                file_path = input()
                send_files(file_path, client_socket)
            elif selection.strip() == '2':
                print("Please enter requested file name:")
                file_name = input()
                req_files(file_name, client_socket)
            elif selection.strip() == '3':
                print("[EXIT] Exit Condition Met\nClosing Client Connection")
                break



    except ConnectionRefusedError:
        print(f"[ERROR] Could not connect to server at {SERVER_IP}:{SERVER_PORT}")

    except Exception as e:
        print(f" Error: {e}")

    finally:
        # Close connection
        client_socket.close()
        print("[DISCONNECT] Successful\n")

if __name__ == "__main__":
    start_client()
```

server.py

```
import socket

import threading

import os


SERVER_IP = '127.0.0.1'

SERVER_PORT = 5001

FILE_STORE = "server_files"


connected_clients = []

client_names = {}


if not os.path.exists(FILE_STORE):

    os.makedirs(FILE_STORE)

    print(f"[CREATED] Directory: {FILE_STORE}")


def handle_client(client_socket, client_address):

    # handles connecting clients

    print(f"[NEW CONNECTION] {client_address} connected")


    try:

        client_name = client_socket.recv(1024).decode('utf-8').strip()

        client_names[client_socket] = client_name


        while True:

            selection = client_socket.recv(1024).decode('utf-8').strip()

            if selection.strip() == '1':

                print(f"[RECIEVE] {client_name} is sending a file\n")
```

```
            file_name = client_socket.recv(1024).decode('utf-8')
            rec_files(file_name, client_socket, client_address, client_name)
        elif selection.strip() == '2':
            print(f"[SEND] {client_name} wants to receive a file\n")
            file_path = client_socket.recv(1024).decode('utf-8')
            send_files(file_path, client_socket, client_name)
        elif selection.strip() == '3':
            print(f"[CLIENT DISCONNECT] {client_name} ({client_address[0]}) disconnected
from the server")
            break
        else:
            print("Invalid Option")
            break

    except Exception as e:
        print(f"[ERROR] {e}")

def send_files(file_path, client_socket, client_name):
    try:
        file_path = os.path.join(FILE_STORE, file_path)

        if not os.path.exists(file_path):
            print(f"[ERROR] File not found: {file_path}")
            return

        file_name = os.path.basename(file_path)
        file_size = os.path.getsize(file_path)
```

```
        #print(f"[SENDING...] {file_name} to {SERVER_IP}")
        #client_socket.sendall(file_name.encode('utf-8'))

        print(f"[SENDING...] {file_size} bytes to {client_name}")
        client_socket.sendall(str(file_size).encode('utf-8'))

        bytes_sent = 0
        with open(file_path, 'rb') as file:
            while bytes_sent < file_size:
                chunk = file.read(4096)
                if not chunk:
                    break

                client_socket.sendall(chunk)
                bytes_sent += len(chunk)
                progress = (bytes_sent / file_size) * 100
                print(f"\rProgress: {progress:.1f}%", end='', flush=True)

            print(f"\rProgress: 100.0%")
            print(f" Sent {bytes_sent:,} bytes\n")

    except Exception as e:
        print(f"[ERROR] Could not send file\n{e}")

def rec_files(file_name, client_socket, client_address, client_name):
    try:

        print(f"[RECEIVING...] {file_name} from {client_name}")
```

```
print(f"\t({client_address[0]}:{client_address[1]})")

file_size = int(client_socket.recv(1024).decode('utf-8'))
print(f"\tSize: {file_size:,} bytes ({file_size/1024:.2f} KB)")

file_path = os.path.join(FILE_STORE, file_name)

bytes_received = 0

with open(file_path, 'wb') as file:
    print(f"   Receiving data...", end='', flush=True)

    while bytes_received < file_size:
        bytes_to_read = min(4096, file_size - bytes_received)
        chunk = client_socket.recv(bytes_to_read)

        if not chunk:
            print("\nNOT CHUNK")
            break

        file.write(chunk)
        bytes_received += len(chunk)
        if bytes_received % 102400 == 0:
            progress = (bytes_received / file_size) * 100
            print(f"\r   Progress: {progress:.1f}%", end='', flush=True)

print(f"\r   Progress: 100.0%")
```

```
        if bytes_received == file_size:
            print(f"[SAVED] {file_path}")
            print(f"[COMPLETE] {bytes_received:,} bytes\n")
        else:
            print(f"[INCOMPLETE] File Transfer Incomplete {bytes_received}/{file_size}")
    except Exception as e:
        print(f"[ERROR] Files not Recieved\n{e}")




def start_server():
    # Handles server
    server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    server_socket.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
    server_socket.bind((SERVER_IP, SERVER_PORT))
    server_socket.listen(5)

    server_socket.settimeout(1.0)

    print("=" * 60)
    print("Multi-Client File Transfer Server Started")
    print("=" * 60)
    print(f"List6ting on {SERVER_IP}:{SERVER_PORT}")
    print(f"Waiting for Client Connections...\n")

    try:
        while True:
            try:
```

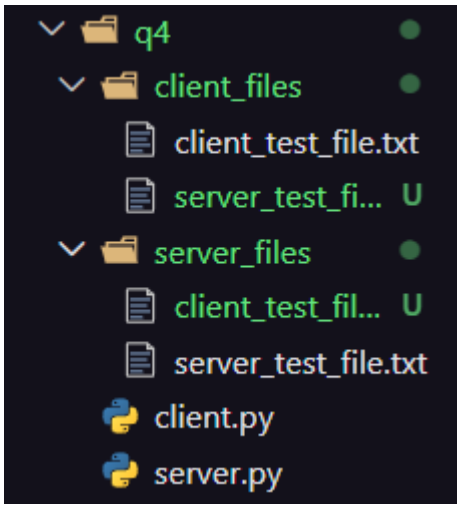
```
            client_socket, client_address = server_socket.accept()
            connected_clients.append(client_socket)

            client_thread = threading.Thread(
                target=handle_client,
                args=(client_socket, client_address)
            )
            client_thread.daemon = True
            client_thread.start()

            # Show Active Connections
            print(f"[ACTIVE CONNECTIONS] {len(connected_clients)} client(s) connected\n")
        except socket.timeout:
            continue
    except KeyboardInterrupt:
        print("\n\n Server Shutting Down")
    finally:
        for clients in connected_clients:
            clients.close()
        server_socket.close()
        print(" Server Stopped")


if __name__ == "__main__":
    start_server()
```

OUTPUT

```
PS C:\Users\lkren\OneDrive\Documents\ITPNA2-T12_Projects\ITPNA2-12_FormativeAssessmen
t_Online_EDUV7579091\q4> py .\server.py
============================================================
Multi-Client File Transfer Server Started
============================================================
List6ting on 127.0.0.1:5001
Waiting for Client Connections...

[NEW CONNECTION] ('127.0.0.1', 53737) connected
[ACTIVE CONNECTIONS] 1 client(s) connected

[RECIEVE] Kyle is sending a file

[RECEIVING...] client_test_file.txt from Kyle
        (127.0.0.1:53737)
        Size: 31 bytes (0.03 KB)
   Progress: 100.0%.
[SAVED] server_files\client_test_file.txt
[COMPLETE] 31 bytes

[SEND] Kyle wants to receive a file

[SENDING...] 31 bytes to Kyle
Progress: 100.0%
 Sent 31 bytes

[CLIENT DISCONNECT] Kyle (127.0.0.1) disconnected from the server


 Server Shutting Down
 Server Stopped
PS C:\Users\lkren\OneDrive\Documents\ITPNA2-T12_Projects\ITPNA2-12_FormativeAssessmen
t_Online_EDUV7579091\q4>
```

```
t_Online_EDUV7579091\q4> py .\client.py
============================================================
Multi-Client File Transfer Server Started
============================================================
Connecting to 127.0.0.1:5001
Connected to Server

Please Enter your Name:

Kyle
Would you like to send or request a file?
 (1) SEND FILE
 (2) REQUEST FILE
 (3) EXIT PROGRAM

1
Please enter file path of the file you want to send:
client_test_file.txt
[SENDING...] client_test_file.txt to 127.0.0.1
[SENDING...] 31 bytes to 127.0.0.1
Progress: 100.0%
 Sent 31 bytes

Would you like to send or request a file?
 (1) SEND FILE
 (2) REQUEST FILE
 (3) EXIT PROGRAM
```

```
2
Please enter requested file name:
server_test_file.txt
[REQUEST] server_test_file.txt from 127.0.0.1
        Size: 31 bytes (0.03 KB)
   Progress: 100.0%.
[SAVED] client_files\server_test_file.txt
[COMPLETE] 31 bytes

Would you like to send or request a file?
 (1) SEND FILE
 (2) REQUEST FILE
 (3) EXIT PROGRAM

3
[EXIT] Exit Condition Met
Closing Client Connection
[DISCONNECT] Successful

PS C:\Users\lkren\OneDrive\Documents\ITPNA2-T12_Projects\ITPNA
t_Online_EDUV7579091\q4>
```



Bibliography

Fadheli, A., 2025. *send-receive-files-using-sockets-python*. [Online]
Available at: https://thepythoncode.com/article/send-receive-files-using-sockets-python
[Accessed 11 March 2026].

Python Software Foundation, 2026. *socket*. [Online]
Available at: https://docs.python.org/3/library/socket.html
[Accessed 10 March 2026].

Python Software Foundation, 2026. *ssl*. [Online]
Available at: https://docs.python.org/3/library/ssl.html
[Accessed 15 March 2026].